FINANCIAL MACHINE LEARNING

The Price Impact of Order Book Events

Replication across Crypto Perpetual and Equity Markets

Project Report

PROJECT TEAM

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Abstract

In this project we reconstruct the empirical framework of Cont, Kukanov, and Stoikov's study "The Price Impact of Order Book Events" and evaluates it in two distinct market environments. In the original paper it is proposed that order flow imbalance (OFI) is a compact measure of net demand and supply at the best bid and ask. Rather than using only executed trades, OFI incorporates market orders, through their observed effect on the top of the limit order book. In this replication we first apply the methodology to five Hyperliquid perpetual markets—BTC, ETH, SOL, HYPE, and DOGE—using three nonconsecutive first-of-month samples in May, June, and July 2026. We then apply the same core quote-based analysis to one regular one-day trading session for AAPL and AMZN.

We start by converting each quote update into an order flow imbalance (OFI) measure. We then group these OFI figures into 10-second intervals and match them with the changes in the mid-price (measured in ticks, i.e., the smallest price unit) that occur over those same intervals. For every 30-minute period, we run a separate regression using Ordinary Least Squares (OLS). To make sure our standard errors are reliable, we apply the White HC0 correction, which adjusts for heteroskedasticity (where the error variance is not constant). Beyond this baseline, we also run extra tests to see if the price impact is non-linear, whether the OFI coefficient tends to fall as market depth rises, and how trade imbalance and trading volume affect the results.

The main finding is very consistent. Across the 720 regressions run on cryptocurrency data (each covering a 30-minute window), every single OFI coefficient was positive. What's more, 99.58% of these coefficients were statistically significant at the 5% level, meaning we can be 95% confident these results aren't just due to chance. Looking at the model's explanatory power (the R^2), the average across the five cryptocurrencies was 0.4406. For comparison, the equity stocks—AAPL and AMZN—gave average R^2 values of 0.5528 and 0.7401 respectively. The simple average for these two stocks comes out to 0.6464, which is very close to the roughly 0.65 benchmark reported in the original paper.

However, crypto results were affected by a drop in quote frequency during July. OFI outperformed both trade imbalance and trading volume, though those two variables added more incremental information in crypto than in the original equity study. Overall, the project confirms that OFI is a useful cross-market measure of short-run price pressure, while also revealing key differences from equities—namely in sampling, nonlinearity, depth sensitivity, and market structure.

Keywords: order flow imbalance; price impact; market depth; limit order book; trade imbalance; Hyperliquid; perpetual futures; NBBO; high-frequency data

Executive Summary of Results

1. In this project we estimate 720 half-hour regressions for the five crypto assets and 26 half-hour regressions for the two equities.
2. Every main-model OFI coefficient is positive. Significance at the 5% level is 99.58% in crypto and 92.31% for each equity.
3. Mean R^2 is 42.52% for BTC, 51.27% for ETH, 45.38% for SOL, 35.92% for HYPE, and 45.19% for DOGE.
4. AAPL and AMZN produce mean R^2 values of 55.28% and 74.01%; the simple average of the two is 64.64%.
5. Quote frequency falls from roughly 12–18 observations per 10-second bin in May and June to approximately 1.8 in July for all crypto assets. The decline coincides with a substantial reduction in R^2.
6. The nonlinear term improves mean R^2 by 2.28–7.69 percentage points across crypto assets. The estimated nonlinear coefficient is usually negative, indicating recurrent concavity over the observed range.
7. The inverse relationship between price impact and depth is strong for SOL, HYPE, DOGE, AAPL, and AMZN, weak for ETH, and contrary to the theoretical sign in the pooled BTC regression.
8. In Section 4, OFI has a higher stand-alone R^2 than trade imbalance for all five crypto assets. Nevertheless, trade imbalance remains significant in 44.44% to 77.08% of the joint half-hour regressions, more often than in the original paper.
9. The estimated price–volume exponent H is well below 0.5 for every crypto asset. Absolute OFI has greater explanatory power than transformed volume, but trading volume retains notable incremental information in the crypto sample.
10. Transaction-price R^2 rises with the number of trades in the price-change horizon. These overlapping trade-time observations create serial dependence, so their reported HC0 p-values require caution.

Project Objective and Scope

The starting point of this project was a practical question: can the relationship between order flow imbalance and short-run price changes be reproduced with datasets that we can collect, clean, and analyze ourself? Instead of rewriting the paper, the goal was to build the full pipeline—from raw quote files to final regressions—and then compare the results across different markets.

The project was completed in two parts. The first part uses Hyperliquid data for five perpetual markets and includes both quote and trade information. The second part uses AAPL and AMZN quote data, which is closer to the stock-market setting of the paper. The same 10-second aggregation and half-hour regression setup is used in both parts, so the results are directly comparable at the methodological level.

Pipeline summary:

- Event-level OFI from changes in best bid, best ask, and their volumes as input.
- 10-second aggregation of OFI, mid-price, spread, depth, and quote counts.
- Separate half-hour OLS regressions with HC0 robust standard errors.
- A nonlinear model using OFI × |OFI|.
- A price-impact versus market-depth regression.
- Trade imbalance, total volume, and transaction-price tests for crypto markets.

Short Summary of the Paper

The paper's main idea is that very short-run price changes are better explained by the net change in supply and demand at the best bid and ask than by executed trades alone. A trade-only variable ignores limit orders and cancellations, even though these events can add or remove a large amount of liquidity from the top of the order book.

To solve this problem, the paper defines order flow imbalance (OFI). Positive OFI means that displayed buying pressure increased or selling pressure decreased. Negative OFI means the opposite. The paper then aggregates OFI over short intervals and estimates a linear relationship between OFI and changes in the mid-price. It also tests whether the price-impact coefficient is larger when the market is less deep.

The original study uses 50 U.S. stocks and reports an average R^2 of about 0.65 for the linear OFI model. It also finds that OFI is stronger than trade imbalance and trading volume. These findings provide the reference point for the implementation in this report.

Main equations used in the implementation

The paper defines e_n as the net change in order flow (demand minus supply) caused by event n:

$$e_n = I_{\{P_n^B \geq P_{n-1}^B\}} q_n^B - I_{\{P_n^B \leq P_{n-1}^B\}} q_{n-1}^B - I_{\{P_n^A \leq P_{n-1}^A\}} q_n^A + I_{\{P_n^A \geq P_{n-1}^A\}} q_{n-1}^A$$

To aggregate these micro-events over a fixed time interval Δt (e.g., 10 seconds):

$$OFI_k = \sum_{n=N(t_{k-1})+1}^{N(t_k)} e_n$$

With the Order Flow Imbalance ($\mathrm{OFI_k}$) defined over the time interval $\Delta\mathrm{t}$, the paper models short-term mid-price movements ($\Delta\mathrm{P_k}$, in ticks) using the linear regression:

$$\Delta P_k = \alpha_i + \beta_i \cdot OFI_k + \epsilon_k$$

Here, β_i represents the price impact coefficient—a parameter inversely proportional to market depth that determines how many ticks the price moves per unit of order imbalance during a 30-minute block i—while α_i is an intercept term (empirically close to zero), and ϵ_k captures residual noise from price rounding or deeper order book levels.

To extend the linear regression framework and test whether price impact has non-linear characteristics, the paper introduces a quadratic term into the regression model:

$$\Delta P_k = \alpha_Q, i + \beta_{Q,i} \cdot OFI_k + \gamma_{Q,i} \cdot OFI_k |OFI_k| + \epsilon_{Q,k}$$

Here, $\gamma_{Q,i}$ measures potential non-linear or concave/convex price impact, where a non-zero value would imply that larger order imbalances move prices disproportionately more or less per share than smaller ones. However, empirical testing across the 50 stocks reveals that adding this non-linear term yields a negligible increase in R^2 (from 65% to 68%) while $\gamma_{Q,i}$ remains statistically insignificant in most time windows, confirming that the relationship between Order Flow Imbalance and short-term price changes is strictly linear.

To build directly on the linear model by showing how the price impact slope β_i changes over time, the paper models its relationship with average market depth (AD_i) in a 30-minute block i using a log-linear regression:

$$\log(\beta_i) = \alpha_{L,i} - \lambda \log(AD_i) + \epsilon_{L,i}$$

Here, λ represents the depth elasticity of price impact, measuring how sensitively price impact decreases as resting order book depth increases, while $\alpha_{L,i}$ acts as an intercept term (representing $\log c$), and $\epsilon_{L,i}$ accounts for residual noise. Empirical estimation across the stock sample shows that $\hat{\lambda}$ is remarkably close to 1 for most stocks, confirming the paper's key theoretical prediction: price impact is inversely proportional to market depth ($\beta_i \propto \frac{1}{AD_i}$).

Trade Imbalance and Trading Volume

$$TI_k = BuyVolume_k - SellVolume_k$$

$$VOL_k = BuyVolume_k + SellVolume_k$$

Trade imbalance records the direction of aggressive transactions, whereas total trading volume records only the amount of activity. The original paper argues that an apparent square-root relation between price changes and trading volume can arise statistically because both volume and OFI increase with the number of underlying events, while the signed sum OFI grows more slowly than unsigned volume.

Data Sources and Data Collection Process

Finding suitable high-frequency order-book data was one of the most difficult parts of this project. The original study uses detailed U.S. equity quote and trade data from the NYSE TAQ database. Access to comparable data is usually restricted, commercially licensed, or available only through institutional databases. In addition, many freely available financial datasets contain only OHLC prices or executed trades and do not include the best bid, best ask, and their associated queue sizes. Since these variables are necessary for constructing order flow imbalance, a large part of the initial work involved searching for a dataset with the correct market-microstructure fields rather than simply finding price data.

Hyperliquid Data

Because a sufficiently detailed and easily accessible equity dataset was not available at the beginning of the project, we decided to test the methodology in a decentralized finance market. Hyperliquid was selected because it operates a central limit order book and provides a market structure that is conceptually similar to conventional electronic exchanges. At each point in time, the best bid price, best ask price, bid size, and ask size can be observed, which makes it possible to construct OFI using almost the same logic as in the original paper.

Hyperliquid also has several practical advantages for an implementation project. It is an active perpetual-futures market, it operates continuously, and its market data can be accessed through public interfaces. Historical Hyperliquid quote and trade data are also distributed by Tardis.dev. Tardis provides free first-of-month sample files for several exchanges and data types, which allowed us to obtain historical top-of-book quotes and executed trades without requiring access to an institutional database. We therefore collected data for BTC, ETH, SOL, HYPE, and DOGE for the first days of May, June, and July 2026.

Using Hyperliquid was not only a solution to the data-access problem. It also created an opportunity to examine whether the main OFI result is specific to U.S. equities or can also be observed in a decentralized perpetual-futures market. The crypto data therefore became an extension of the original study rather than simply a substitute for the unavailable equity data. The availability of both quote and trade files also allowed us to implement the paper’s comparisons between OFI, trade imbalance, and trading volume.

Crypto sample

Item	Project setup
Markets	BTC, ETH, SOL, HYPE, DOGE
Dates	1 May, 1 June, and 1 July 2026
Quote variables	Best bid/ask price and size
Trade variables	Taker side, price, and amount
Time aggregation	10 seconds
Regression window	30 minutes

Equity Data

After completing the first version of the analysis on Hyperliquid, we continued searching for data that were closer to the market studied in the original paper. We eventually found a sample equity dataset that had been provided alongside a commercially listed dataset. The full dataset was being offered for sale, but the available sample contained a large number of high-frequency quote observations for AAPL and AMZN. The file included the variables required for the core quote-based analysis: timestamps, best bid prices, best ask prices, bid amounts, and ask amounts.

The equity sample contained approximately 2.7 million quote observations in total. After restricting the data to the regular U.S. trading session and removing invalid, locked, or crossed quotes, approximately 803,000 valid observations remained for AAPL and 1.86 million for AMZN. This was sufficient to construct event-level OFI, aggregate the data into 10-second intervals, and estimate 13 separate half-hour regressions for each stock. The equity file did not include the corresponding transaction records. For this reason, it was used for the core OFI, nonlinear-impact, market-depth, and intraday analyses, while the trade-imbalance and trading-volume tests were conducted using the Hyperliquid data. Despite containing only two stocks and one trading session, the sample was valuable because its structure was much closer to the quote data used in the original paper. It also provided a useful benchmark for comparing the behavior of OFI in a traditional equity market with its behavior in decentralized cryptocurrency markets.

Overall, the data-collection process developed in two stages. Hyperliquid was first selected because detailed order-book and trade data were practically accessible and suitable for reproducing the methodology. The later discovery of the AAPL and AMZN quote sample then allowed us to test the same implementation in a market environment closer to the original study. Combining the two datasets made it possible to evaluate both the reproducibility of the original equity result and its broader applicability across different market structures.

Equity sample

Item	Project setup
Stocks	AAPL and AMZN
Session	One regular trading session, 09:30–16:00
Variables	Best bid/ask price and size
Rows after processing	AAPL: 803,356; AMZN: 1,855,226
10-second intervals	2,340 per stock
Available analyses	Core OFI, nonlinear, depth, intraday

Implementation in Python

The code is divided into preparation files and analysis files. The preparation stage converts raw market data into a common format. The analysis stage reads the processed 10-second files and runs the same regression functions for each asset. This separation made the project easier to debug and also makes it possible to use the code with another dataset later.

Data cleaning and continuous segments

For every quote file, invalid prices and sizes are removed, timestamps are sorted, and exact duplicate quote states are dropped. A new segment is started after a data gap longer than 15 seconds. The first quote after a gap is not compared with the last quote before the gap, which avoids creating artificial OFI values.

Code excerpt 1. Event-level OFI calculation.

```
previous_bid_price = data.groupby(group_columns)["best_bid_price"].shift(1)
previous_bid_size  = data.groupby(group_columns)["best_bid_size"].shift(1)
previous_ask_price = data.groupby(group_columns)["best_ask_price"].shift(1)
previous_ask_size  = data.groupby(group_columns)["best_ask_size"].shift(1)

bid_positive = (data["best_bid_price"] >= previous_bid_price).astype(float) * data["best_bid_size"]
bid_negative = (data["best_bid_price"] <= previous_bid_price).astype(float) * previous_bid_size
ask_negative = (data["best_ask_price"] <= previous_ask_price).astype(float) * data["best_ask_size"]
ask_positive = (data["best_ask_price"] >= previous_ask_price).astype(float) * previous_ask_size

data["event_ofi"] = bid_positive - bid_negative - ask_negative + ask_positive
```

This is the most important part of the implementation. An increase in the bid or a reduction in the ask contributes positively. A decrease in the bid or an increase in the ask contributes negatively. The formula handles both price changes and changes in queue size at an unchanged price.

Building the 10-second dataset

Code excerpt 2. Ten-second aggregation.

```
grouped = data.groupby([
    "source_date",
    pd.Grouper(key="timestamp_utc", freq="10s"),
]).agg(
    OFI=("event_ofi", lambda x: x.sum(min_count=1)),
    ending_mid_price=("mid_price", "last"),
    average_depth=("depth", "mean"),
    average_spread=("spread", "mean"),
    quote_count=("timestamp_utc", "size"),
).reset_index()

grouped["delta_mid_ticks"] = (
    grouped["ending_mid_price"].diff() / tick_size
)
```

OFI is calculated before aggregation, because calculating it only from the first and last quote of a 10-second interval would miss the events inside the interval. Mid-price changes are divided by tick size so that the dependent variable is measured in ticks.

Half-hour regressions

Code excerpt 3. Main OFI regression.

```
y = regression_data["delta_mid_ticks"]
X = sm.add_constant(regression_data[["OFI"]], has_constant="add")

linear_model = sm.OLS(y, X).fit(cov_type="HC0")

beta = linear_model.params["OFI"]
r_squared = linear_model.rsquared
p_value = linear_model.pvalues["OFI"]
```

A separate model is estimated for each half-hour. A full half-hour contains up to 180 ten-second observations, and the code requires at least 120 usable observations. HC0 robust standard errors are used because financial high-frequency residuals usually have changing variance.

Nonlinear and depth tests

Code excerpt 4. Nonlinear OFI model.

```
regression_data["ofi_nonlinear"] = (
    regression_data["OFI"] * regression_data["OFI"].abs()
)

X_nl = sm.add_constant(
    regression_data[["OFI", "ofi_nonlinear"]],
    has_constant="add",
)
nonlinear_model = sm.OLS(y, X_nl).fit(cov_type="HC0")
```

The nonlinear model is compared with the linear model using the change in R^2 and the sign of γ. For the depth test, only positive β estimates are used because the model is estimated in logarithms. HAC standard errors are used for this second-stage regression.

Trade variables for Section 4

Code excerpt 5. Signed trade volume and trade imbalance.

```
data["signed_trade_amount"] = np.where(
    data["trade_side"] == "buy",
    data["trade_amount"],
    -data["trade_amount"],
)

trade_imbalance = signed_trade_amount.sum()
trade_volume = trade_amount.sum()
```

Because the trade files contain the taker side, buyer-initiated trades receive a positive sign and seller-initiated trades receive a negative sign. The signed amounts are summed to create trade imbalance, while unsigned amounts are summed to create total volume.

Special timestamp handling for the stock file

Code excerpt 6. Decoding the stock time-of-day timestamp.

```
nanosecond = values % 1_000_000_000
hhmmss = values // 1_000_000_000

second = hhmmss % 100
minute = (hhmmss // 100) % 100
hour = hhmmss // 10_000

time_nanoseconds = (
    (hour * 3600 + minute * 60 + second) * 1_000_000_000
    + nanosecond
)
```

This parser was necessary because the stock file does not store a standard Unix timestamp. After decoding the hour, minute, second, and nanosecond fields, only regular-session observations are retained.

Data Quality and Descriptive Statistics

Crypto Data Quality

Table 4. Processed crypto data summary

Asset	Quote rows	Trade rows	10-second bins	Tick	Mean quotes/bin	Bins with trades
BTC	319,880	1,571,845	25,920	1	12.34	100.0%
ETH	313,954	503,020	25,918	0.1	12.11	98.9%
SOL	267,759	340,689	25,918	0.001	10.33	94.9%
HYPE	294,821	1,398,875	25,919	0.001	11.37	100.0%
DOGE	233,859	66,303	25,853	1e-05	9.05	63.9%

All five markets contain approximately 25,900 non-empty 10-second bins across the three dates. Transaction intensity differs substantially: BTC and HYPE contain more than 1.3 million trades, whereas DOGE contains roughly 66 thousand. Such heterogeneity can affect trade-time regressions even when the clock-time OFI analysis remains comparable.

Quote-Sampling Heterogeneity across Dates

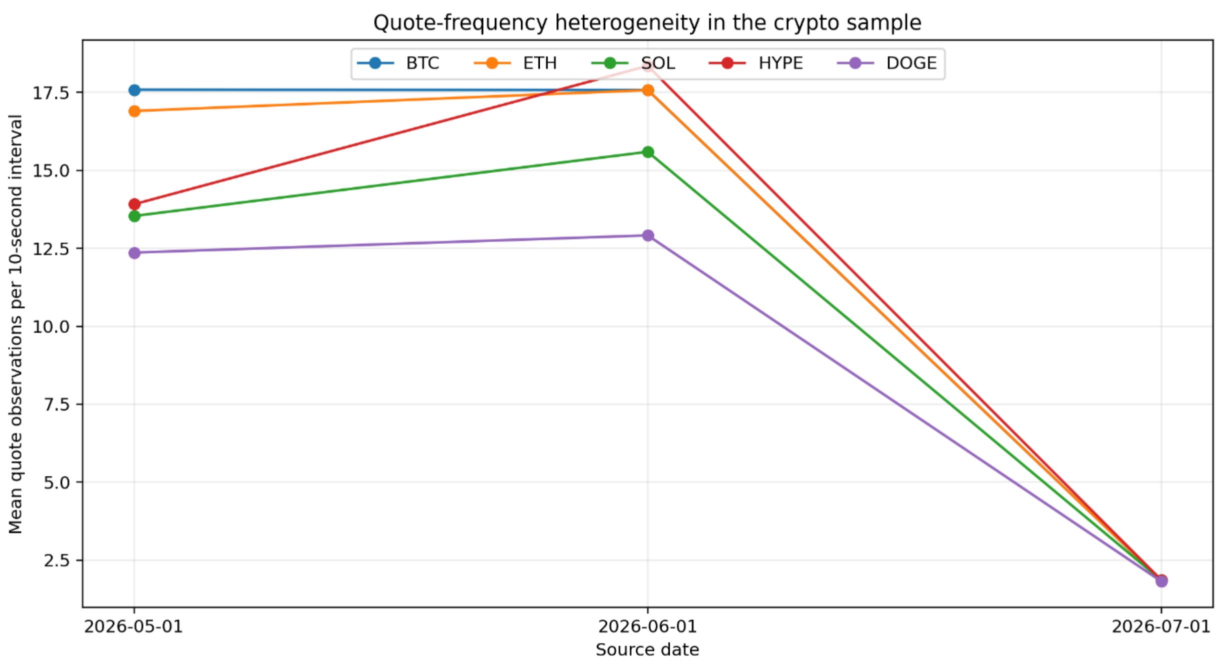

Figure 2. Mean quote count per 10-second bin. July represents a common low-frequency sampling regime across all five crypto assets.

Table 5. Quote frequency and mean half-hour R^2 by source date

Asset	May quotes	June quotes	July quotes	May R^2	June R^2	July R^2
BTC	17.59	17.58	1.86	0.508	0.457	0.311
ETH	16.91	17.58	1.86	0.586	0.609	0.343
SOL	13.54	15.60	1.86	0.542	0.525	0.294
HYPE	13.91	18.35	1.86	0.419	0.414	0.245
DOGE	12.36	12.92	1.82	0.550	0.558	0.248

May and June usually contain 12–18 quote observations per 10-second bin, while July contains approximately 1.82–1.86. Every asset experiences a lower mean R^2 in July. A plausible mechanical explanation is measurement error: at lower sampling frequency, multiple order-book transitions occur between observed snapshots and are compressed into a single net change.

Equity Data Quality

Table 6. AAPL and AMZN data-quality summary

Symbol	Raw rows	Regular-session rows	Valid rows	Invalid rows removed	Tick	10-second bins	Mean quotes/bin
AAPL	816,373	805,925	803,356	2,569	0.01	2,340	343.3
AMZN	1,878,771	1,862,387	1,855,226	7,161	0.01	2,340	792.8

Each equity contains exactly 2,340 ten-second bins, corresponding to 6.5 regular-session hours. Quote density is much higher than in the crypto sample: approximately 343 updates per bin for AAPL and 793 for AMZN. This is one plausible reason the equity estimates are closer to the original paper's explanatory power.

Main Results for Crypto Perpetual Markets

Linear OFI Model

Table 7. Half-hour linear OFI results for crypto markets

Asset	N	Mean R^2	Median R^2	$\beta > 0$	Significant β	Mean $t(\beta)$
BTC	144	0.425	0.437	100.0%	100.0%	8.90
ETH	144	0.513	0.549	100.0%	100.0%	10.46
SOL	144	0.454	0.478	100.0%	100.0%	10.26
HYPE	144	0.359	0.369	100.0%	97.9%	6.99
DOGE	144	0.452	0.491	100.0%	100.0%	9.83

The results are very stable: all 720 β estimates are positive, with only 3 insignificant at the 5% level (99.58% significance). ETH has the highest mean R^2, while HYPE has the lowest. This consistent sign and significance provide strong evidence that the OFI mechanism generalizes to crypto.

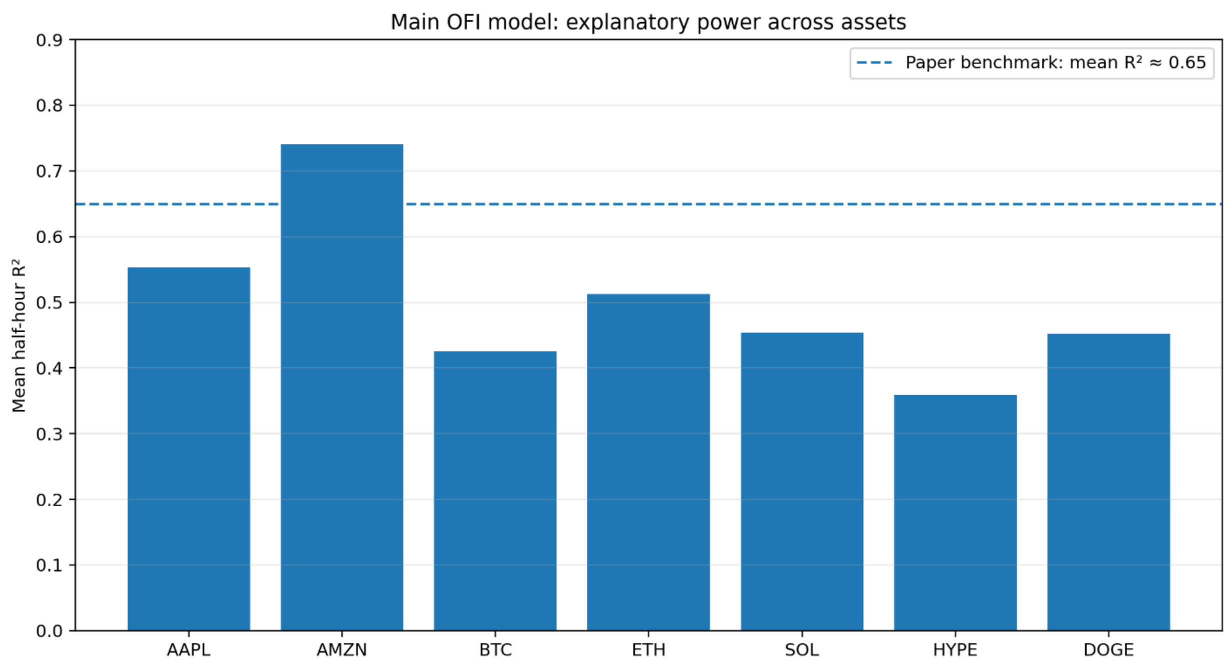

Figure 3. Mean explanatory power of the principal OFI model across the seven assets. The dashed line is the original paper's approximate 65% benchmark.

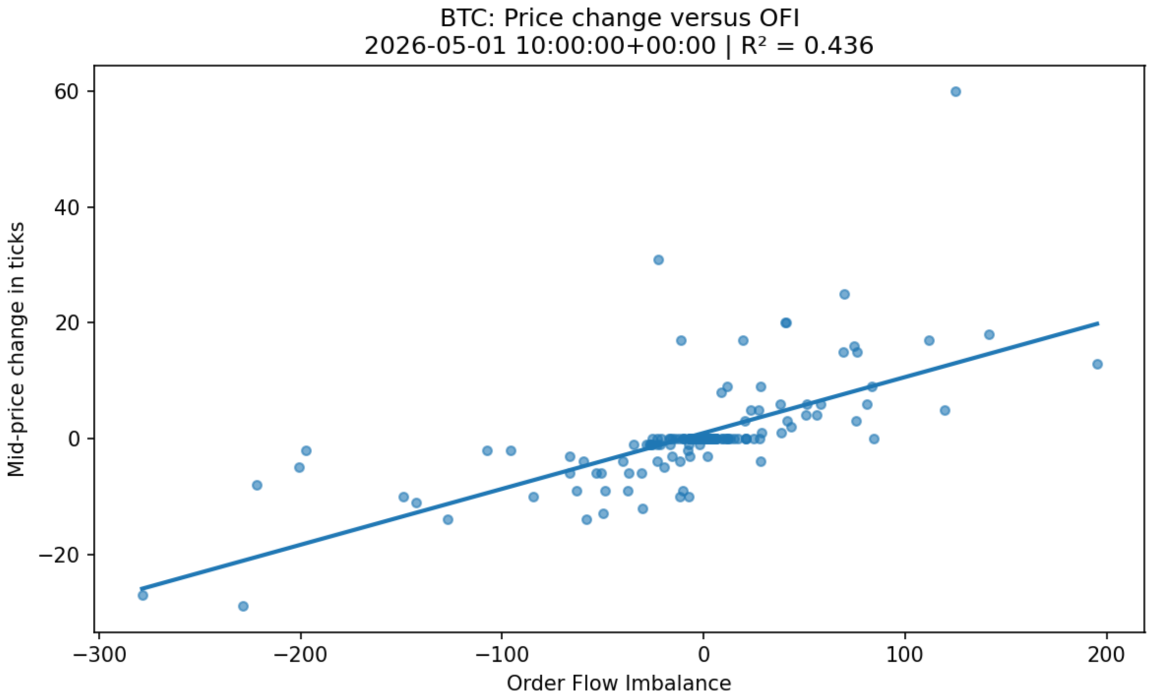

Figure 4. Representative BTC half-hour regression. The positive slope captures the contemporaneous relationship between OFI and the change in mid-price.

Effect of the July Sampling Regime

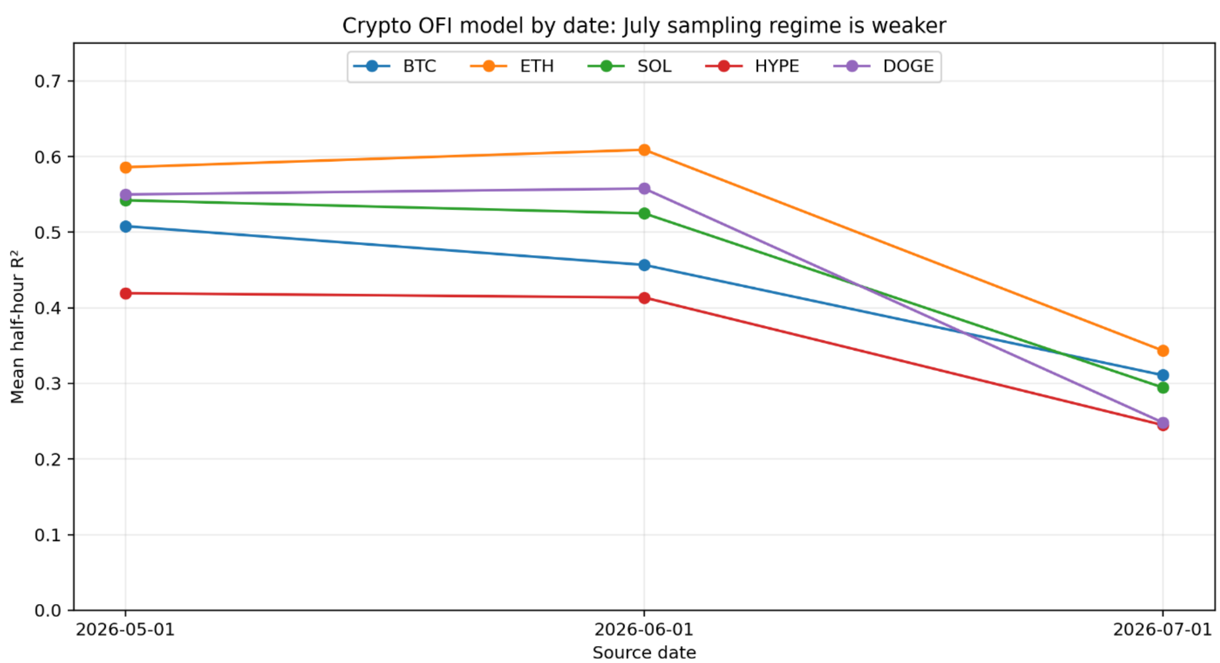

Figure 5. Mean crypto R^2 by source date. The common July decline coincides with the decline in quote frequency.

In May–June, ETH, SOL, and DOGE had mean R^2 values around 0.52–0.61, but in July these fell to about 0.25–0.34. So the overall crypto average of 44% blends two distinct regimes. The key takeaway: the OFI relationship doesn't vanish in July—coefficients stay positive and significant—but sparser quote data simply explain less of the price variation.'

Nonlinear Model

Table 8. Nonlinear OFI × |OFI| results in crypto

Asset	Mean gain (pp)	Median gain (pp)	Significant γ	γ < 0	γ < 0 and significant
BTC	3.14	2.52	74.3%	97.2%	74.3%
ETH	2.34	1.54	55.6%	89.6%	54.9%
SOL	2.28	1.18	50.0%	95.8%	49.3%
HYPE	7.69	6.42	84.0%	97.9%	84.0%
DOGE	4.03	1.44	55.6%	88.9%	55.6%

Mean R^2 improvements range from 2.28 percentage points for SOL to 7.69 for HYPE. The estimated γ is negative in approximately 89% to 98% of half-hours, so the nonlinear evidence primarily takes the form of concavity. The linear model nevertheless remains the appropriate benchmark because the incremental gain is modest for most assets and the nonlinear effect is market-dependent.

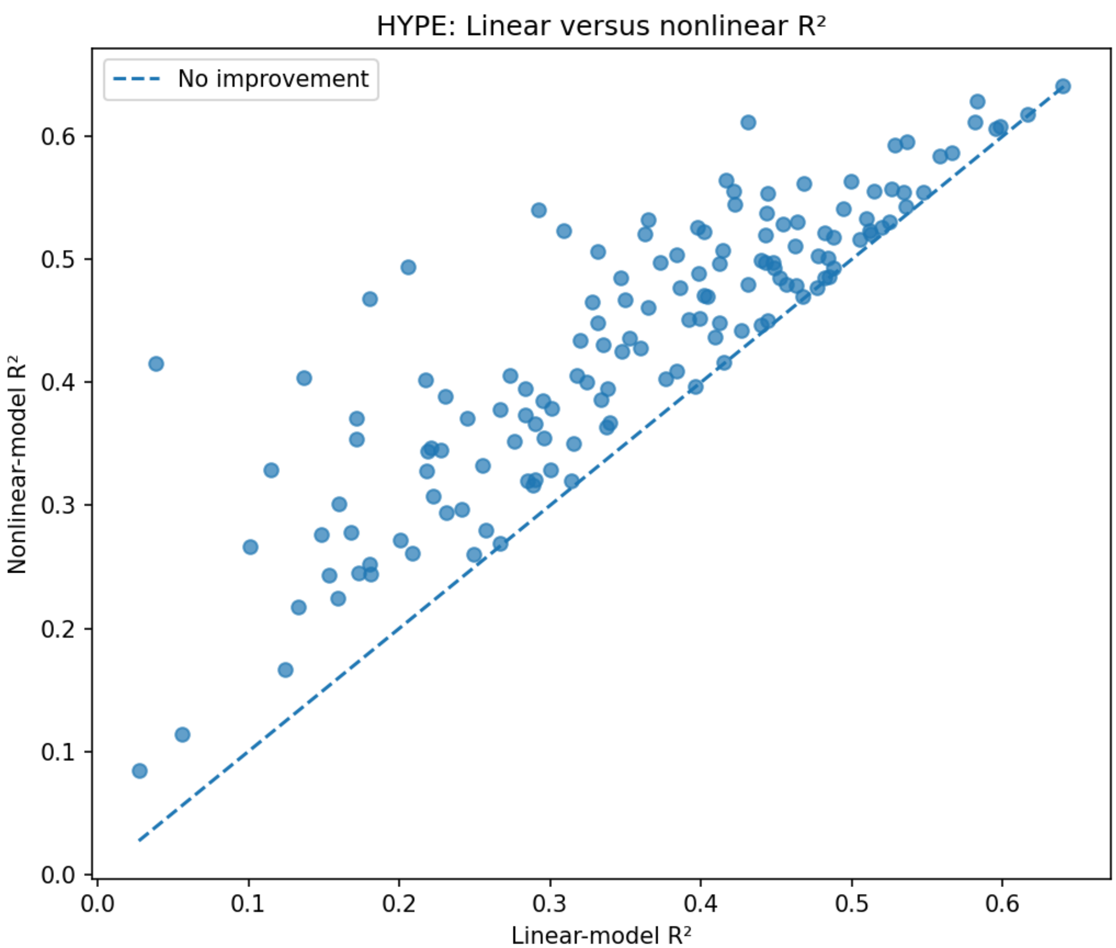

Figure 6. HYPE linear versus nonlinear R^2. Its points lie farther above the 45-degree line than those of the other crypto assets, making HYPE the clearest nonlinear case.

Heavy-Tailed Residuals

Table 9. Excess kurtosis of crypto linear-model residuals

Asset	Median excess kurtosis	Mean	Maximum
BTC	6.39	11.22	112.63
ETH	6.77	12.61	92.09
SOL	5.98	12.97	150.55
HYPE	3.16	5.97	75.94
DOGE	3.17	8.06	101.63

Excess kurtosis is often positive and can be extreme (SOL's maximum exceeds 150), pointing to heavy tails and potential outlier influence. HC0 adjusts for heteroskedasticity, but it does not protect against outliers or serial correlation in the residuals. As a result, very high t-statistics should not be taken at face value—they may be inflated by a few extreme observations or by time-series dependence. It's therefore advisable to interpret them alongside residual diagnostics and checks for autocorrelation to avoid overstating significance.

Price Impact and Market Depth

Table 10. Pooled price-impact/depth relationship in crypto

Asset	Pooled λ	Log–log R^2	Corr(β, depth)	p-value: H_0 $\lambda = 1$
BTC	-0.601	0.045	0.171	0.000
ETH	0.215	0.026	-0.129	0.000
SOL	1.186	0.167	-0.365	0.484
HYPE	1.322	0.522	-0.161	0.131
DOGE	0.604	0.479	-0.521	0.000

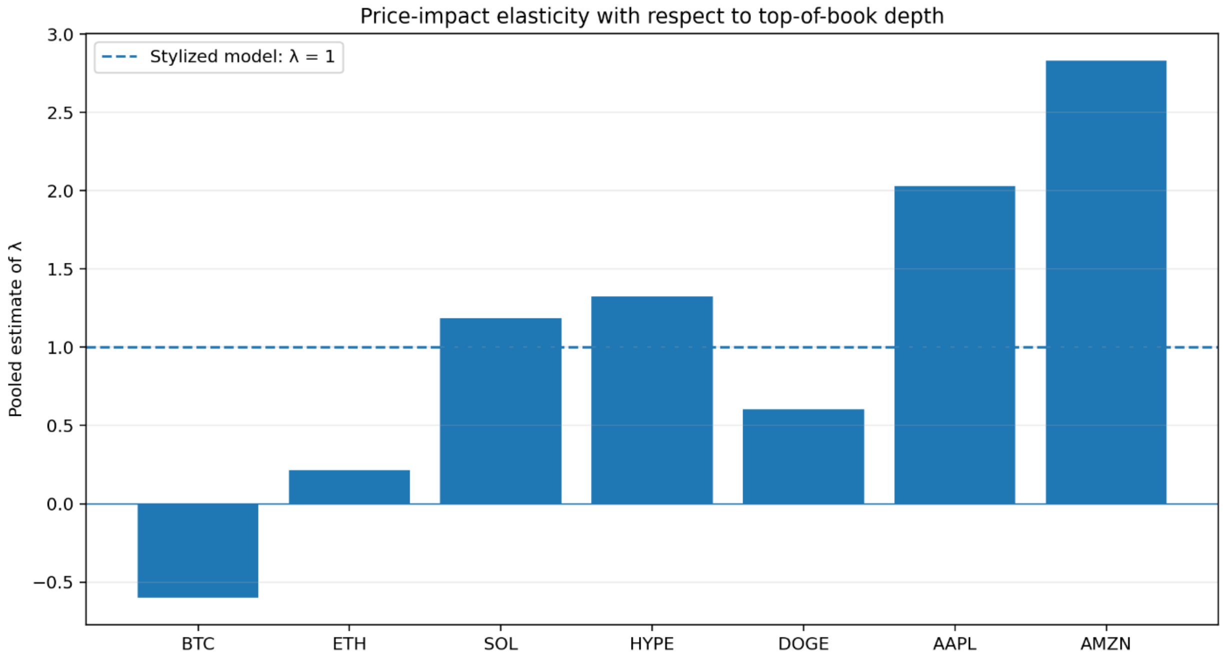

Figure 7. Pooled depth-elasticity estimates for crypto and equity assets. The dashed λ = 1 line is the simplified theoretical benchmark.

SOL, HYPE, and DOGE display the clearest inverse relationship. HYPE has a log–log R^2 of approximately 0.52 and DOGE approximately 0.48. ETH has a weak inverse relation. The pooled BTC estimate is negative, contrary to the theoretical sign. Because the depth regression combines three quote-sampling regimes, especially the low-frequency July sample, the pooled λ estimates should be treated as descriptive and regime-sensitive rather than universal structural elasticities.

Section 4 Results for the Crypto Sample

OFI versus Trade Imbalance

Table 11. OFI and trade-imbalance comparison

Asset	OFI R^2	TI R^2	Joint R^2	OFI significant in joint	TI significant in joint
BTC	0.425	0.131	0.505	100.0%	75.0%
ETH	0.513	0.076	0.555	100.0%	53.5%
SOL	0.454	0.069	0.491	100.0%	44.4%
HYPE	0.359	0.189	0.439	96.5%	77.1%
DOGE	0.452	0.048	0.474	99.3%	45.8%

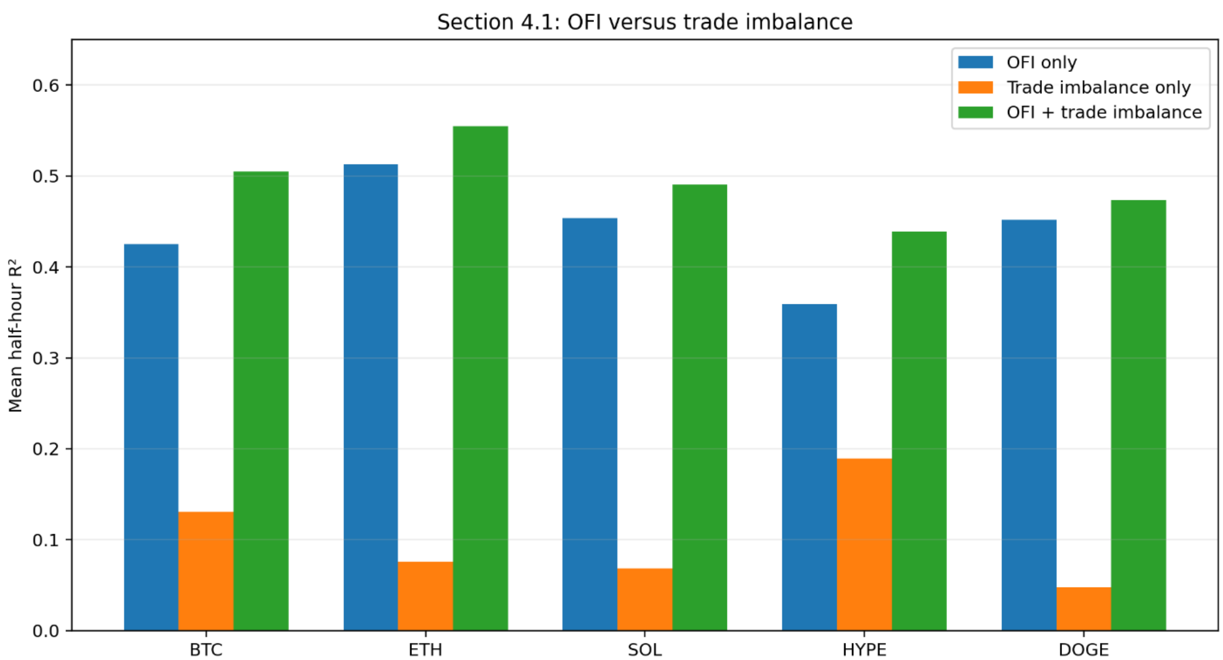

Figure 8. OFI has greater stand-alone explanatory power than trade imbalance for all five assets; the joint model has the largest R^2.

The first result replicates clearly: OFI consistently outperforms trade imbalance (TI) across all markets—for ETH, mean R^2 jumps from 7.57% (TI) to 51.27% (OFI). The second result, however, does not fully replicate: after controlling for OFI, TI remains significant in 44–77% of crypto half-hours, compared with roughly 31% in the original paper. Possible explanations include the single-venue setup, direct taker-side information (rather than inferred trade classification), perpetual market features, and differences in quote–trade synchronization. Still, statistical significance should be distinguished from economic contribution—the joint-model R^2 gain from adding TI is actually smaller than the high significance rates might imply.

Trading Volume and the Estimated Exponent H

Table 12. Absolute OFI versus transformed trading volume

Asset	Mean H	$\lvert OFI \rvert$ R^2	$Volume^H$ R^2	Joint R^2	$\lvert OFI \rvert$ significant	$Volume^H$ significant
BTC	0.149	0.237	0.156	0.343	94.4%	98.6%
ETH	0.111	0.319	0.136	0.395	97.2%	86.1%
SOL	0.091	0.232	0.080	0.286	91.0%	75.7%
HYPE	0.201	0.148	0.138	0.237	72.2%	81.9%
DOGE	0.063	0.245	0.051	0.275	84.7%	34.7%

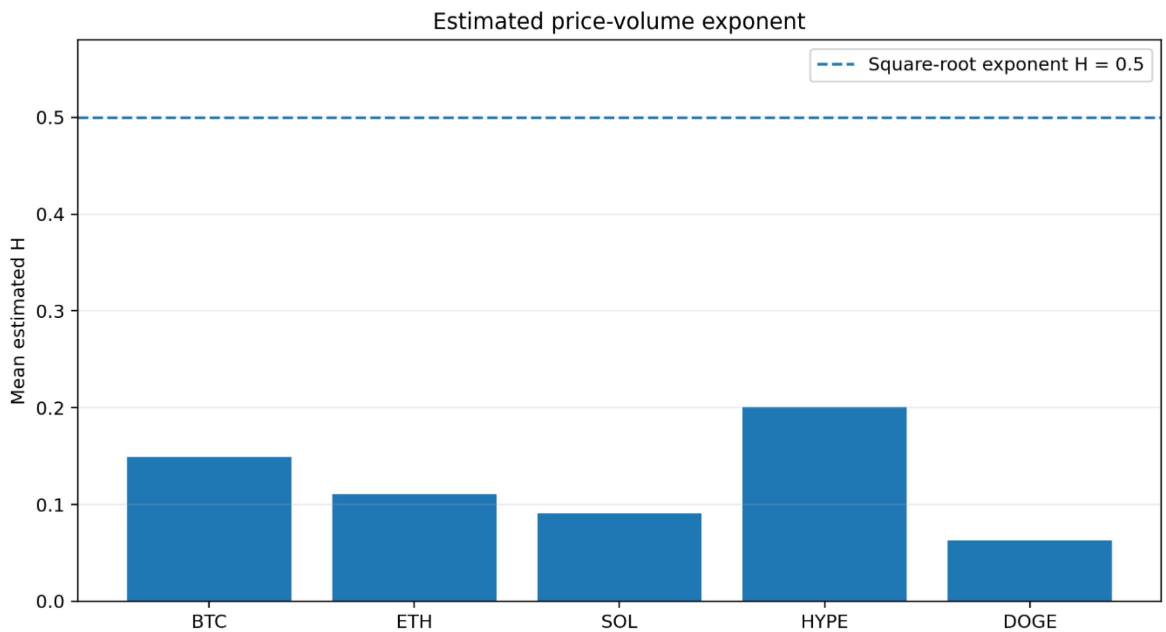

Figure 9. Mean price–volume exponent H. Every estimate is well below the square-root benchmark of 0.5.

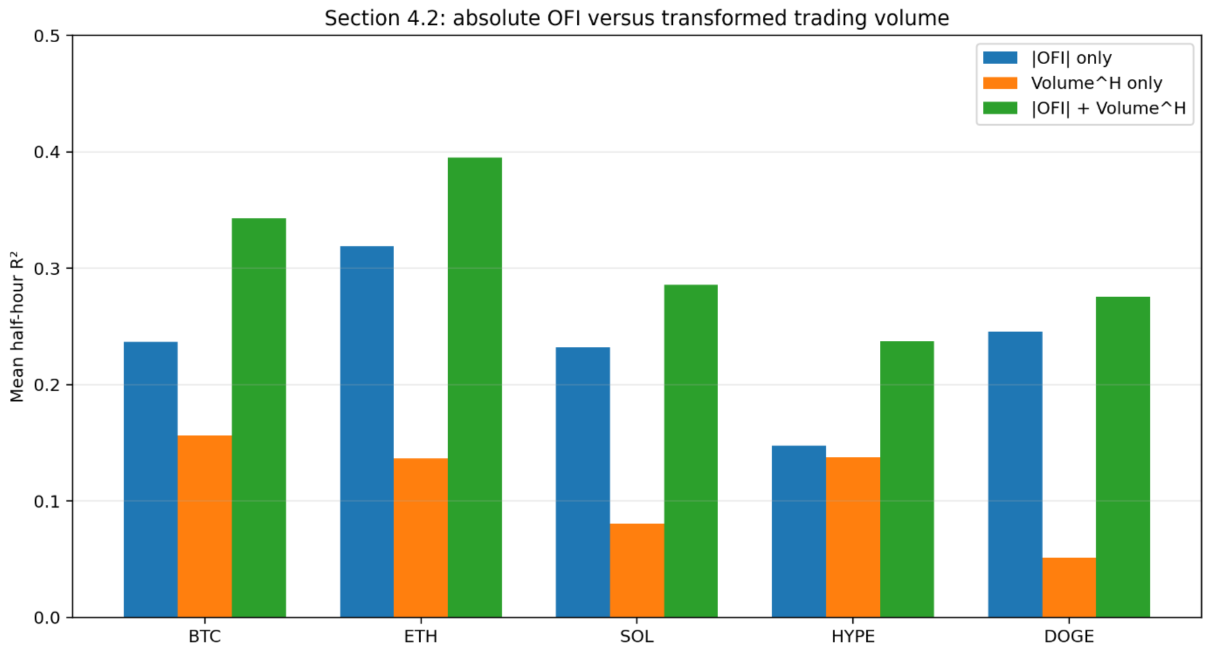

Figure 10. Absolute OFI has greater stand-alone R^2 than transformed volume for every asset, while the joint model improves explanatory power.

Mean H ranges from 0.063 for DOGE to 0.201 for HYPE, and the estimates are far below the fixed square-root exponent. This supports the original paper's criticism of a stable square-root law. Absolute OFI has greater explanatory power than transformed volume in all five markets. Unlike the original paper, however, volume remains significant in 34.72% to 98.61% of the joint regressions. Trading activity therefore contains more independent information in this crypto sample than in the paper's equity sample.

Transaction-Price Robustness

Table 13. Pooled transaction-price results at a 10-trade horizon

Asset	OFI R^2 (L = 10)	TI R^2 (L = 10)	Joint R^2 (L = 10)
BTC	0.074	0.076	0.147
ETH	0.283	0.037	0.315
SOL	0.229	0.032	0.255
HYPE	0.033	0.133	0.157
DOGE	0.265	0.036	0.293

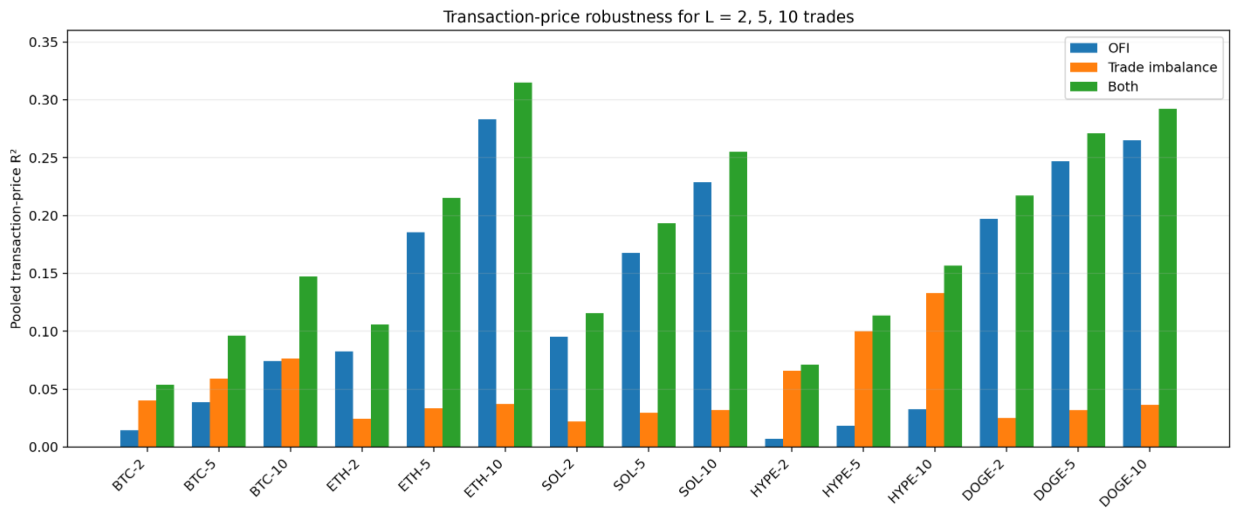

Figure 11. Transaction-price robustness for L = 2, 5, and 10 trades. R^2 increases with horizon; OFI is clearly stronger for ETH, SOL, and DOGE, while HYPE is an exception.

ETH, SOL, and DOGE continue to favor OFI over TI in trade time. HYPE strongly favors TI, and BTC produces similar stand-alone R^2 values for the two variables at L = 10. A common result is that R^2 increases with the number of trades in the horizon. These pooled L-trade observations overlap heavily and contain very large samples, so their p-values are likely more optimistic than those from a model that explicitly accounts for serial correlation.

Results for AAPL and AMZN

Main Linear Model

Table 14. Main OFI results for the equity sample

Symbol	N	Mean R^2	Median R^2	Minimum	Maximum	Significant β
AAPL	13	0.553	0.583	0.012	0.782	92.3%
AMZN	13	0.740	0.801	0.174	0.877	92.3%

AAPL has a mean R^2 of 55.28% and AMZN 74.01%. The simple average is 64.64%, numerically close to the approximately 65% average in the original paper. The comparison is encouraging, but it is not a formal replication of the paper's cross-sectional estimate because the project includes only two equities and one undated session.

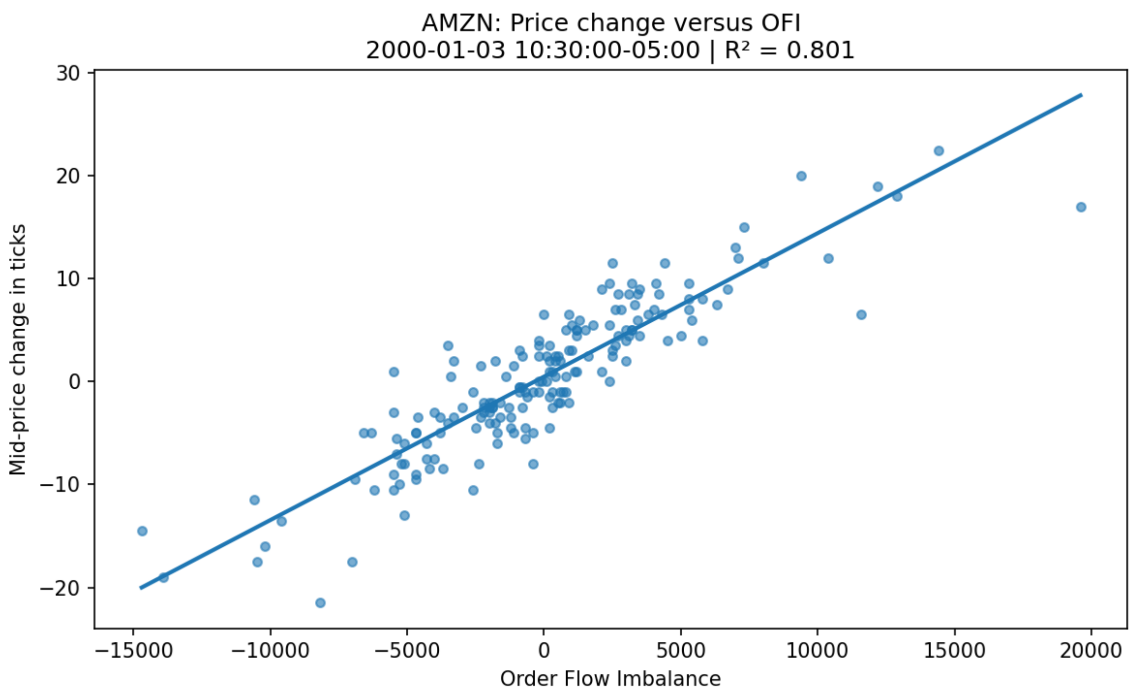

Figure 12. Representative AMZN half-hour regression with R^2 of approximately 0.80. The date in the chart title is the synthetic computational date, not the actual trading date.

Intraday Behavior

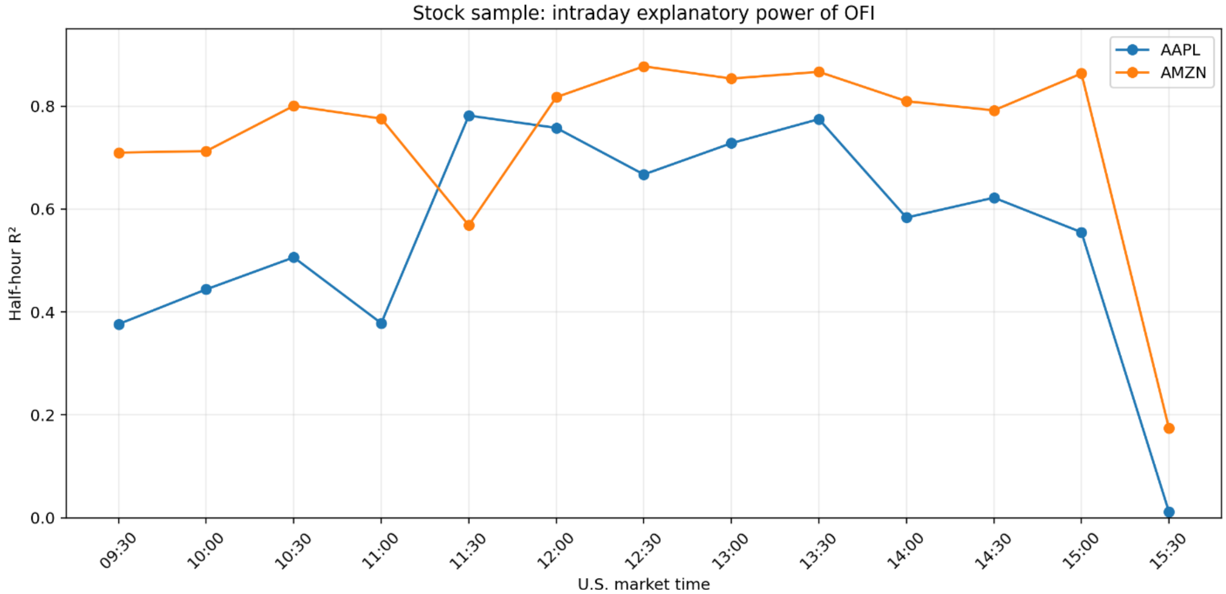

Figure 13. Half-hour explanatory power for AAPL and AMZN. Both stocks exhibit a sharp decline in the final half-hour.

For AAPL, the first twelve half-hours generally produce R^2 values between approximately 0.38 and 0.78, while the 15:30–16:00 interval falls to 0.012 and β becomes insignificant. AMZN produces approximately 0.71–0.88 for most of the session but declines to 0.174 in the final half-hour. The closing-period observation is retained because it is a genuine feature of the data rather than an error to be removed.

Nonlinearity and Heavy Tails

Table 15. Nonlinear results for AAPL and AMZN

Symbol	Mean gain (pp)	Median gain (pp)	Significant γ	γ < 0
AAPL	8.01	1.13	38.5%	53.8%
AMZN	5.38	1.67	53.8%	92.3%

The mean R^2 gains are 8.01 percentage points for AAPL and 5.38 for AMZN, but the median gains are only 1.13 and 1.67. A small number of half-hours therefore raise the mean substantially. The result remains consistent with treating the linear model as the main benchmark while acknowledging occasional nonlinear episodes.

Market Depth

Table 16. Equity depth results

Symbol	λ	Log–log R^2	Corr(β, depth)	p-value: H_0 λ = 1
AAPL	2.032	0.241	-0.516	0.205
AMZN	2.832	0.765	-0.927	0.016

Both stocks have the theoretically expected inverse sign. AAPL has $\lambda \approx 2.03$, but with only 13 observations the null $\lambda = 1$ is not rejected. AMZN has $\lambda \approx 2.83$, log–log $R^2 \approx 0.77$, and rejects $\lambda = 1$. The direction of the relationship is strong, but the exact power estimate is highly sample-sensitive.

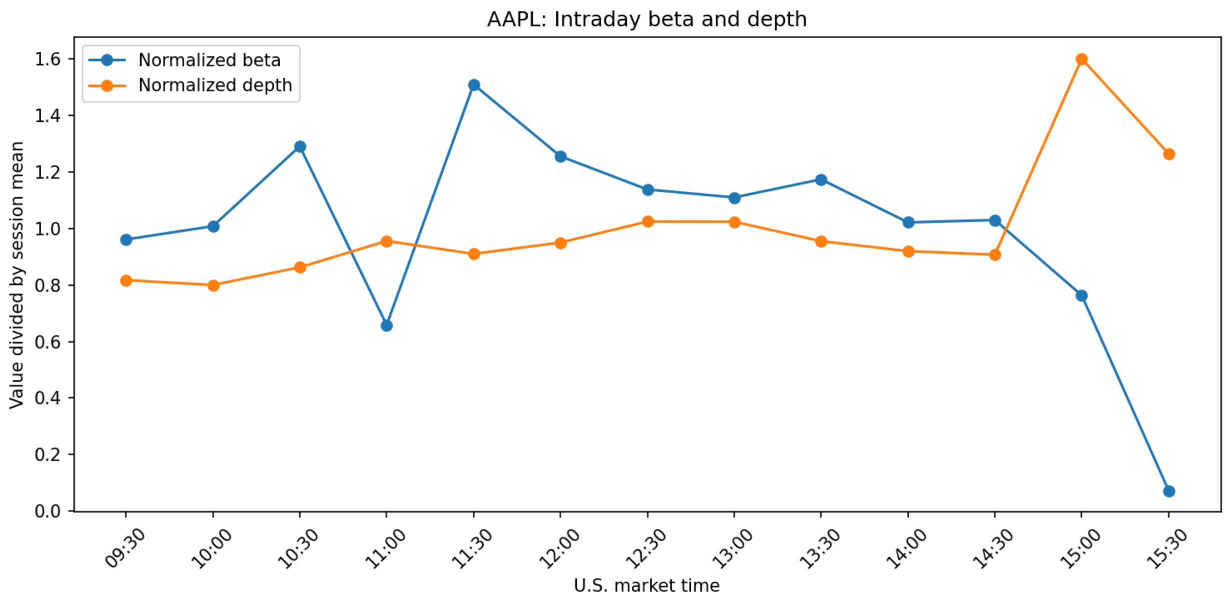

Figure 14. AAPL intraday β and depth profiles. With one session only, the pattern is exploratory rather than a structural seasonality estimate.

Cross-Market Comparison

Table 17. Cross-market summary

Sample	Assets	Mean R^2	β significance	Nonlinearity	Depth result	Section 4 result
Original paper	50 U.S. equities	≈0.650	High	Mostly linear	Inverse; λ close to 1	OFI dominates TI and volume
Project equity sample	AAPL and AMZN	0.646	92.3%	Mean affected by a few intervals	Inverse; λ > 1, small sample	Section 4 not available
Project crypto sample	Five perpetual markets	0.441	99.6%	More recurrent concavity	Heterogeneous; inverse sign in 4 of 5 pooled estimates	OFI stronger; TI and volume add information

The equity sample is closer to the paper in explanatory power, while crypto preserves exceptionally strong stability in the sign and significance of β but leaves a larger share of short-run price variation unexplained by observed OFI. The difference in quote density is substantial: the equity file contains hundreds of updates per 10-second bin, compared with roughly 12–18 in the higher-frequency crypto dates and approximately 1.8 in July.

Nonlinearity is more recurrent in crypto, especially for HYPE. In equities, the difference between mean and median R^2 improvements indicates that a few intervals drive the average nonlinear gain. Trade imbalance and volume also preserve more incremental information in the crypto environment. These differences are consistent with a single-venue market, 24/7 trading, perpetual-contract mechanics, and heterogeneous quote capture.

Cross-market interpretation

The evidence supports the general economic mechanism—net top-of-book demand and supply is closely associated with short-run prices—without implying that a single coefficient, elasticity, or R^2 should be universal across asset classes or data feeds.

Final Discussion and Conclusion

Main findings

1. The complete data pipeline was successfully implemented for both crypto and equity quote data.
2. The OFI coefficient is positive in all 746 main half-hour regressions: 720 crypto regressions and 26 equity regressions.
3. The main model explains a meaningful part of very short-run price changes in every asset.
4. OFI is stronger than trade imbalance and transformed volume when each variable is used alone.
5. The nonlinear model improves the fit, especially for HYPE, but the linear model remains a clear and useful benchmark.
6. Price impact generally falls when top-of-book depth increases, although the strength of this relation varies across assets.
7. The AAPL and AMZN results are particularly close to the original paper's reported explanatory power.

What the project shows

The most important lesson from the implementation is that short-run prices respond to changes in the whole top-of-book state, not only to completed trades. Limit-order additions and cancellations matter because they directly change the amount of displayed supply and demand. OFI combines these events into one variable and produces a stable relationship with price changes across very different markets.

The project also shows why it is useful to test a model in more than one market. The stock sample gives results close to the paper, while the crypto sample shows that the same idea remains useful in a 24/7 perpetual market even though trade imbalance, volume, and nonlinearity sometimes add more information.

Conclusion

The project achieved its main objective. The paper's central OFI model was translated into reusable Python code, applied to seven assets, and extended to trade imbalance and volume for the crypto sample. The results are both statistically strong and economically intuitive. Across all assets, positive net order-book pressure is associated with positive short-run price changes, and the relationship is especially strong in the equity sample. The final conclusion is that OFI is a practical and informative summary of top-of-book supply and demand, and the model transfers well from the original stock-market setting to crypto perpetual markets.

Appendix: File Structure and Execution Order

The project is organised into separate configuration, preparation, analysis, and comparison files.

File	Purpose
config.py	Crypto paths and settings
prepare_data.py	Clean quotes and build 10-second OFI data
analyze.py	Run core crypto regressions and figures
prepare_trades.py	Prepare crypto trade variables
analyze_section4.py	Run Section 4 tests
stock_config.py	Stock paths and settings
prepare_stock_data.py	Parse and prepare stock quotes
analyze_stocks.py	Run stock regressions and figures
compare_markets.py	Compare stock and crypto summaries

Table A1. Main Python files in the submitted project.

Execution order

```
python prepare_data.py
python analyze.py
python prepare_trades.py
python analyze_section4.py
python prepare_stock_data.py
python analyze_stocks.py
python compare_markets.py
```

The preparation scripts create the processed CSV and Parquet files. The analysis scripts then create asset-level tables, figures, and cross-market summaries. To apply the project to another dataset, the main changes are the file paths, column mapping, and timestamp parser; the regression functions can remain the same.